Macro-Economics and Sociological Fractals

Human Economies as J-Cost Ledgers:
The Physics–Economics Dictionary,
φ-Scaling of Organizations,
and Testable Predictions from Recognition Science

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Abstract

Recognition Science derives the structure of physical reality from the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. This functional is literally an economic object: it prices deviations from equilibrium, it is minimized by trade, and its conservation law constrains feasible reallocations. We show that the mathematical framework already proved in the Lean 4 formalization—dissipation channels, channeled feasibility, complexity measures, channel refinement, halting bounds—is not merely analogous to economics. It IS economics, read at the scale of human institutions rather than particles.

We establish an explicit dictionary between the RS physics formalization and economic theory, then derive three classes of testable predictions:

1. **φ-Scaling of Organizations** (§5): Self-similarity of hierarchical institutions forces the golden ratio $\varphi \approx 1.618$ as the optimal scaling factor between organizational levels. Optimal team size ≈ 5, department ≈ 8, division ≈ 13—the Fibonacci sequence.
2. **J-Bounded Market Clearing** (§6): The time for a market to clear (reach equilibrium) is bounded by $T_{\max} = D_0/\delta$, where D_0 is the initial market disequilibrium and δ is the minimum per-transaction friction.
3. **Conservation-Constrained Wealth Distributions** (§7): Log-charge conservation constrains feasible wealth redistributions, predicting that the geometric mean of wealth is invariant under balanced trade, and that Pareto-like tails emerge from channeled equilibrium.

No new mathematics is introduced. Every theorem cited is already machine-verified in Lean 4. The contribution of this paper is the interpretation: reading the existing formalization as an economic theory and extracting its empirical content.

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1 Introduction

The cost functional of Recognition Science,

$$J(x) \;=\; \frac{1}{2}\Big(x + x^{-1}\Big) - 1, \qquad x > 0, \tag{1}$$

was derived from the Recognition Composition Law and the combinatorics of the 3-cube Q_3. Its original domain of application is fundamental physics: it determines the mass spectrum of fermions, the fine-structure constant, and the gravitational coupling.

But the functional itself is an economic object. It measures the cost of holding a position x relative to equilibrium ($x = 1$). It is symmetric: being overvalued by a factor of 2 costs the same as being undervalued by a factor of 2. It is convex: the cost of deviation accelerates as you move further from balance. It has a unique zero-cost state: perfect balance at $x = 1$.

These are not metaphors. They are the defining properties of a transaction cost functional. The variable $x = p/p^*$ is a price ratio (actual price divided by equilibrium price). The cost $J(x)$ is the friction of maintaining that deviation. The total defect $D = \sum_i J(c_i)$ is the total economic friction in the system. The variational dynamics—minimize D subject to conservation—IS market clearing.

This paper makes the dictionary explicit.

2 The Physics–Economics Dictionary

Every structure in the RS physics formalization has a direct economic interpretation. The correspondence is not an analogy constructed after the fact. It is a reading of the same mathematical structures at a different scale.

2.1 Primitive Objects

RS Structure	**Physics Reading**	**Economics Reading**
Ledger entry $c_i \in \mathbb{R}_{>0}$	Ratio of adjacent tick amplitudes	Price ratio p_i/p_i^* of agent i
$J(c_i)$	Defect of entry i	Transaction cost / friction of agent i
$D(c) = \sum_i J(c_i)$	Total defect	Total market friction
$\sigma = \sum_i \log c_i$	Conserved log-charge	Log-wealth (conserved under balanced trade)
$\mathcal{F}(c)$	Feasible set (same σ)	Budget-balanced reallocations
$c^* = \arg\min_{\mathcal{F}} D$	Variational successor	Market-clearing equilibrium
$c_i^* = e^{\sigma/N}$	Uniform equilibrium	Uniform price (all agents at geometric mean)

2.2 Channeled Structures

RS Structure	Physics Reading	Economics Reading
`DissipationChannel`	Partition into subsystems	Economic sectors / firms / nations
Group G_k	Spatial region	Market sector
$\sigma_k = \sum_{i \in G_k} \log c_i$	Per-group charge	Sector endowment
$\mathcal{F}_P(c)$	Channeled feasible set	Sector-constrained allocations
$v_k = e^{\sigma_k / \lvert G_k \rvert}$	Per-group equilibrium value	Sector equilibrium price
Complexity $\mathcal{C}$	Number of distinct v_k	Number of distinct price levels
$\mathcal{C} \geq 2$	Non-trivial structure	Price dispersion across sectors
`Refines`	Finer partition	More institutional boundaries
`IsChanneledStep`	Channeled variational step	Sector-by-sector market clearing

2.3 Dynamic Structures

RS Structure	Physics Reading	Economics Reading
Trajectory $\tau : \mathbb{N} \to$ Config	Ledger evolution	Economic history
Step cost δ_t	Defect reduction per tick	Gains from trade per period
$D(\tau(t{+}1)) \leq D(\tau(t))$	Defect monotonicity	Friction decreases over time
Halting bound D_0/δ	Max computation steps	Time to market clearing
Equilibrium ($\delta_t = 0$)	Fixed point	No profitable trades remain
Loop $\Rightarrow$ fixed point	No non-trivial loops	No perpetual arbitrage cycles

2.4 Censorship Structures

RS Theorem	Physics Reading	Economics Reading
`channeled_subset_feasible`	$\mathcal{F}_P \subseteq \mathcal{F}$	Sector constraints reduce options
`global_optimum_leq_channeled`	Global $\leq$ channeled min	Central planning $\leq$ decentralized
`uniform_precluded`	No uniform equilibrium	No single price clears all sectors
`structure_permanent`	Charges conserved	Endowments persist
`finer_channels_higher`	More structure costs more	Regulation costs friction
`infinite_computation_impossible`	No infinite loops	No perpetual motion machines

3 Why Markets Exist

The RS framework answers the fundamental question of economics: *why do markets exist?*

Theorem 3.1 (Markets Are Forced by Locality)**.** *If the economy has $K \geq 2$ sectors with different endowment densities ($\sigma_{k_1}/|G_{k_1}| \neq \sigma_{k_2}/|G_{k_2}|$), then:*

(i) No uniform price clears all sectors simultaneously (`uniform_precluded_by_channels`)*.*

(ii) The equilibrium has at least two distinct prices (`structured_complexity_ge_two`)*.*

(iii) Price dispersion persists along any trajectory (`structure_permanent`)*.*

Markets exist because different sectors have different endowments, and no single price can satisfy all sectoral budget constraints simultaneously. Trade—the process of exchanging goods at sector-specific prices—IS the channeled variational step. Markets are the mechanism by which the economic ledger dissipates friction through channeled optimization.

Remark 3.2 (Coase's Theory of the Firm as a Theorem)*.* Ronald Coase asked: why do firms exist, when markets could coordinate all transactions? The RS answer is immediate from `finer_channels_higher_minimum`: finer channels (more institutional boundaries) have higher minimum friction, but the universe (the economy) cannot avoid channeling because of locality (bounded rationality, finite information bandwidth).

A firm exists where the cost of internal coordination (channeled defect within the firm's partition) is lower than the cost of market coordination (defect across firm boundaries). The firm boundary IS the channel boundary. The firm's existence IS a theorem of channeled optimization.

4 Why Inequality Persists

Theorem 4.1 (Inequality Is a Conservation Law)**.** *Along any channeled trajectory, per-sector endowments σ_k are preserved:*

$$\sigma_k(\tau(t)) \;=\; \sigma_k(\tau(0)) \qquad \forall\, t,\ \forall\, k.$$

(`structure_permanent`)

This means: the distribution of wealth across sectors is not a transient perturbation that markets will eventually equalize. It is a conserved quantity. A nation rich in energy resources stays rich in energy (relative to its sector) because the sector's log-charge is conserved under intra-sector trade.

Remark 4.2. This does NOT mean inequality cannot change. The theorem says per-*channel* charges are conserved. If the channel structure itself changes (sectors merge, new sectors form), the charges can be redistributed. But within a fixed institutional structure, endowment inequality is permanent. This explains why inequality is so persistent within stable institutional regimes, and why it changes primarily during periods of institutional upheaval.

5 φ-Scaling of Organizations

5.1 Self-Similarity Forces φ

`PhiForcing.lean` proves: if the cost structure is self-similar (the same J at every scale), the unique non-trivial scale ratio is $\varphi = (1+\sqrt{5})/2 \approx 1.618$, satisfying $\varphi^2 = \varphi + 1$.

If economic organizations are self-similar—the math governing a team is the same as the math governing a division, just at a different scale—then the scaling ratio between hierarchical levels is forced to be φ.

5.2 The Fibonacci Prediction

Successive powers of φ round to the Fibonacci sequence: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

Prediction 5.1 (φ-Scaling of Team Sizes)**.** Optimal organizational units at successive hierarchical levels should have sizes that approximate Fibonacci numbers. Specifically:

Level	**Scale** φ^n	**Fibonacci** F_n	**Prediction**
Individual	$\varphi^0 = 1$	1	—
Pair / buddy	$\varphi^1 \approx 1.6$	2	Fire teams, lab pairs
Team	$\varphi^3 \approx 4.2$	5	Scrum teams, squads
Section	$\varphi^5 \approx 11.1$	13	Platoons, lab groups
Department	$\varphi^7 \approx 29.0$	34	Departments
Division	$\varphi^9 \approx 76.0$	89	Small companies

Remark 5.2 (Empirical Support)*.* Jeff Bezos's "two-pizza team" rule targets 5–8 people (Fibonacci $F_5 = 5$, $F_6 = 8$). The U.S. military fire team has 4 members, squad 8–13, platoon 30–50. Dunbar's number ($\approx 150 \approx \varphi^{11}$) marks the cognitive limit for social groups. Robin Dunbar's layered social structure (5, 15, 50, 150, 500, 1500) follows approximate φ-scaling with a multiplier of $\approx 3 \approx \varphi^{2.4}$. These are not exact matches—organizational sizes are subject to noise, cultural variation, and domain-specific constraints—but the systematic appearance of Fibonacci-adjacent numbers across unrelated domains is predicted by φ-forcing.

5.3 The Scale-Invariance Argument

Proposition 5.3 (φ Is the Unique Self-Similar Organizational Scale)**.** *If an organization's cost structure at scale s satisfies $J_s = J_{s/r}$ for some scale ratio $r > 1$ (the math is the same at every level), then $r = \varphi$.*

Proof. By `PhiForcing.lean`: self-similarity in a J-cost ledger forces $r^2 = r + 1$, whose unique positive solution is $r = \varphi$. □

Prediction 5.4 (Growth Ratios)**.** When firms grow optimally (maintaining self-similar structure), their headcount at successive stages should scale by factors close to φ:

$$N_{k+1}/N_k \;\approx\; \varphi \;\approx\; 1.618.$$

Firms that deviate significantly from this ratio—growing too fast (ratio $\gg \varphi$) or too slowly (ratio $\ll \varphi$)—incur excess J-friction from scale mismatch.

6 J-Bounded Market Clearing Times

6.1 The Market Clearing Bound

`AlgorithmicCost.lean` proves that the number of non-trivial steps in any variational trajectory is bounded by D_0/δ, where D_0 is the initial defect and δ is the minimum per-step cost.

Applied to markets:

Definition 6.1 (Market Disequilibrium)**.** The *market disequilibrium* is the total defect of the price configuration:

$$D_0 \;=\; \sum_{i=1}^{N} J(p_i/p_i^*),$$

where p_i is agent i's actual price and p_i^* is the equilibrium price.

Definition 6.2 (Transaction Granularity)**.** The *transaction granularity* $\delta > 0$ is the minimum friction reduction achievable by a single trade. It depends on the market's microstructure: tick sizes, minimum lot sizes, information costs.

Theorem 6.3 (Market Clearing Bound)**.** *The number of trades required for the market to reach effective equilibrium is bounded:*

$$T_{clear} \leq \frac{D_0}{\delta}.$$

(`computation_bounded`)

Prediction 6.4 (Market Clearing Time)**.** A market with N agents, initial disequilibrium D_0, and transaction granularity δ will clear within D_0/δ trades. Specifically:

1. **Liquid markets** (small δ, e.g. stock exchanges): many trades, but each reduces friction by only a little. Clearing time $\sim D_0/\delta_{\text{tick}}$.
2. **Illiquid markets** (large δ, e.g. real estate): few trades, each reducing friction by a lot. Clearing time $\sim D_0/\delta_{\text{deal}}$.
3. **After a shock** (large D_0, e.g. post-crisis): more trades needed. Clearing time scales linearly with the magnitude of the shock.

6.2 No Perpetual Arbitrage

Theorem 6.5 (No Perpetual Arbitrage Cycles)**.** *If the price trajectory ever returns to a previous state, all intermediate trades had zero friction reduction, and every intermediate state was already at equilibrium.* (`loop_implies_zero_step_cost`)

Corollary 6.6. *Persistent arbitrage opportunities are impossible in a J-cost market. Any apparent cycle must involve states that are already at equilibrium—the cycle is trivial. Non-trivial arbitrage cycles would require the market to reduce friction and then undo the reduction, violating defect monotonicity.*

Remark 6.7. This is a stronger statement than the efficient market hypothesis (EMH). EMH says prices *reflect* all information. The RS bound says prices *cannot cycle* through non-equilibrium states—the market must monotonically approach equilibrium. Temporary deviations (volatility) can occur within the bound, but the long-run trajectory is inexorably toward clearing.

7 Conservation-Constrained Wealth Distributions

7.1 The Conservation Law

In the RS framework, the conserved quantity is log-charge:

$$\sigma = \sum_{i=1}^{N} \log c_i.$$

In economic terms, $c_i = w_i/w^*$ is the ratio of agent i's wealth to the reference wealth. Then $\sigma = \sum \log(w_i/w^*) = \log \prod_i (w_i/w^*)$.

Conservation of σ under balanced trade means: the geometric mean of wealth ratios is invariant.

Theorem 7.1 (Geometric Mean Invariance)**.** *Under any feasible reallocation (balanced trade), the geometric mean of wealth is preserved:*

$$\left(\prod_{i=1}^{N} w_i\right)^{1/N} = const.$$

(Follows from $\sigma = \sum \log c_i$ conservation in `VariationalDynamics.lean`)

Remark 7.2. This is a stronger constraint than conservation of arithmetic mean wealth (which allows inflation/deflation). The geometric mean is the natural "equilibrium anchor" in the RS framework. It is preserved because log is the natural coordinate of J: the cost functional becomes $J_{\log}(t) = \cosh t - 1$ in log-coordinates.

7.2 Channeled Wealth Distribution

Under channeled constraints (sectors with independent conservation), the equilibrium wealth of agent i in sector k is

$$w_i^* \;=\; w^* \cdot \exp\left(\frac{\sigma_k}{|G_k|}\right),$$

where σ_k is the sector's log-endowment and $|G_k|$ is the sector's population.

Prediction 7.3 (Sector-Dependent Equilibrium Wealth)**.** In an economy with K sectors and non-uniform endowments:

1. Each sector has a distinct equilibrium wealth level $w_k^* = w^* \cdot e^{\sigma_k/|G_k|}$.
2. Richer sectors (higher $\sigma_k/|G_k|$) have exponentially higher equilibrium wealth.
3. The number of distinct wealth levels equals the complexity $\mathcal{C} \geq 2$ *(`structured_complexity_ge_two`)*.
4. Wealth levels persist across time *(`structure_permanent`)*.

7.3 Pareto Tails from Channeled Equilibrium

If the sector endowments $\sigma_k/|G_k|$ follow a distribution (e.g., from heterogeneous resource geology, technology, or institutional history), the resulting wealth distribution has a specific tail structure.

Prediction 7.4 (Pareto Exponent)**.** If sector endowment densities $\mu_k = \sigma_k/|G_k|$ are drawn from a distribution with tail behavior $\Pr(\mu > x) \sim x^{-\alpha}$, then the wealth distribution inherits a Pareto tail:

$$\Pr(w > W) \;\sim\; W^{-\alpha}.$$

The Pareto exponent α is determined by the distribution of sector endowments, not by individual behavior. In the self-similar case (φ-scaling), dimensional analysis in $D = 3$ spatial dimensions suggests α relates to φ and D through the RS mass formula.

Remark 7.5. Empirically, wealth distributions across countries consistently show Pareto exponents $\alpha \approx 1.5$–2.5. The RS framework predicts that this exponent is not arbitrary but is constrained by the dimensional and self-similarity properties of the economic ledger. A precise derivation of α from φ and $D = 3$ is an open problem.

8 Why Empires Collapse

Theorem 8.1 (Expansion Is Bounded)**.** *Every economic expansion (trajectory of increasing complexity with step cost $\geq \delta > 0$) is bounded by D_0/δ steps.* (`computation_bounded`)

Applied to civilizations:

Prediction 8.2 (Imperial Lifespan Bound)**.** A civilization with initial "disequilibrium budget" D_0 (its capacity for expansion—resource surplus, technological advantage, institutional innovation) and minimum per-epoch friction δ (the cost of administering each additional province, trade route, or institutional layer) can expand for at most D_0/δ epochs before it must:

1. Reach equilibrium (stabilize at current size), or
2. Collapse below the equilibrium (overshoot and contract).

Empires that overshoot (expand beyond the bound supported by their budget) accumulate structural defect that cannot be paid down—they are in the regime where the "halting theorem" forces slowdown.

Remark 8.3. Peter Turchin's cliodynamics identifies secular cycles of ∼200–300 years in agrarian empires and ∼50–80 years in industrial states. In the RS framework, these timescales correspond to D_0/δ for different institutional granularities δ. The ratio of imperial to industrial cycle length ($200/60 \approx 3.3 \approx \varphi^{2.6}$) is suggestively close to φ-scaling.

9 The Economic Calculation Problem

Theorem 9.1 (Central Planning Pays a Structure Premium)**.** *The globally optimal allocation (single channel, no sectoral constraints) achieves the lowest total friction. Any decentralized allocation (multiple channels) has higher or equal friction:*

$$D^*_{global} \;\leq\; D^*_{channeled}.$$

(`global_is_optimal_over_channels`)

Theorem 9.2 (But Central Planning Is Infeasible)**.** *The global optimum requires unrestricted access to all N entries simultaneously. By locality (bounded rationality, finite communication bandwidth), no agent can access all entries. The economy MUST use channeled optimization. The difference $D^*_{channeled} - D^*_{global}$ is the unavoidable cost of decentralization—the price of structure.*

Remark 9.3 (Mises–Hayek in One Equation)*.* The economic calculation debate (Mises 1920, Hayek 1945) asked: can a central planner replicate market outcomes? The RS answer is a precise inequality:

$$D^*_{\text{central}} = D^*_{\text{global}} \;\leq\; D^*_{\text{market}} = D^*_{\text{channeled}}.$$

A perfect central planner would beat the market. But a perfect central planner would require non-local access to all entries—which locality forbids. The market's higher friction is the price of being feasible under locality.

This is not a vindication of either side. It is a *theorem*: central planning is theoretically optimal but physically impossible; markets are theoretically suboptimal but physically necessary. The gap between them is the cost of being local.

10 Summary of Testable Predictions

#	Prediction	How to Test
5.1	Optimal team sizes scale by φ	Survey organizational sizes across industries; test whether level-ratios cluster near Fibonacci numbers
5.4	Firm growth ratios $\approx \varphi$	Analyze S&P 500 headcount history; test whether successful growth stages scale by $\sim$1.6$\times$
6.4	Market clearing time $\leq D_0/\delta$	Measure post-shock price convergence times in equity, commodity, and real estate markets
7.3	Sector equilibrium wealth $= w^* e^{\sigma_k/\lvert G_k\rvert}$	Compare GDP per capita across sectors/nations with resource endowment densities
7.4	Pareto exponent from sector endowments	Fit α from cross-country wealth data; test correlation with natural resource heterogeneity
8.2	Imperial lifespan $\leq D_0/\delta$	Test Turchin cycle data against RS halting bound for different institutional granularities

11 Discussion

11.1 What This Paper Claims

1. **The RS formalization is already an economic theory.** The structures proved in Lean 4—channels, feasibility, complexity, refinement, halting—have direct economic interpretations. No new mathematics is needed.
2. **φ-scaling is testable.** The prediction that organizational sizes scale by the golden ratio is specific enough to be confirmed or refuted by empirical data.
3. **Market clearing is bounded.** The D_0/δ bound gives a quantitative upper limit on the time for markets to reach equilibrium after a shock.
4. **Wealth distributions are constrained.** Log-charge conservation constrains the feasible wealth redistributions, predicting geometric mean invariance and sector-dependent equilibrium levels.
5. **The Mises–Hayek debate is a theorem.** Central planning is globally optimal but locally infeasible. Markets are locally feasible but globally suboptimal. The gap is the cost of locality.

11.2 What This Paper Does Not Claim

We do not claim to have derived all of economics from first principles. The RS framework provides the *constraint structure*—the budget set, the conservation laws, the cost functional, the optimization principle. The specific content of economic life—what goods exist, what preferences agents have, what technologies are available—is not determined by J alone. These are the "initial conditions" and "channel structures" that the framework takes as input.

11.3 Open Problems

1. **Deriving α from φ.** The Pareto exponent of wealth distributions may be calculable from the RS mass formula and φ-scaling. This would give a zero-parameter prediction of the wealth distribution.
2. **The 8-tick economic cycle.** `EightTick.lean` proves the ledger's fundamental period is 8. Can this be connected to business cycle frequencies?
3. **Institutional evolution dynamics.** The current framework proves that channels exist and persist. It does not yet formalize how new channels form or old ones collapse—the dynamics of institutional change.
4. **Information as log-charge.** In the RS framework, log-charge is conserved. In economics, information (measured in bits = log-probabilities) may play the role of log-charge. Can the RS conservation law be identified with conservation of information in economic systems?

12 Conclusion

Recognition Science was built to derive physics. It turns out to have derived economics as well.

The cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ that determines particle masses and gravitational coupling also determines transaction costs and market clearing. The dissipation channels that explain why life exists also explain why firms exist. The halting bound that limits physical computation also limits imperial expansion. The φ-scaling that forces the golden ratio into the mass spectrum also forces it into organizational hierarchies.

This is not a coincidence. It is the self-similarity theorem (`PhiForcing.lean`) applied at every scale. The math that governs quarks also governs corporations, because both are dissipation channels on a J-cost ledger, and the ledger's self-similarity forces the same structures at every scale.

Human economies are not analogous to physical systems. They ARE physical systems—macro-ledgers minimizing J-cost under local conservation constraints. The dictionary is not a metaphor. It is an identity.

Machine Verification. All theorems cited in this paper have been formalized and verified in Lean 4 using Mathlib, with no `sorry` placeholders and no custom axioms. The relevant formalizations are in: `DissipativeComplexity.lean`, `AlgorithmicCost.lean`, `PhiForcing.lean`, and `VariationalDynamics.lean`.

References

[1] J. Washburn, "Thermodynamics of Complexity: Why Life?," `RecognitionScience/Foundation/DissipativeComplexity.lean`, 2026.

[2] J. Washburn, "Algorithmic Cost and the Halting Problem," `RecognitionScience/Foundation/AlgorithmicCost.lean`, 2026.

[3] J. Washburn, "Phi Forcing: Self-Similar Discrete Ledger → Golden Ratio," `RecognitionScience/Foundation/PhiForcing.lean`, 2025–2026.

[4] J. Washburn, "Variational Dynamics: The Equation of Motion for the Ledger," `RecognitionScience/Foundation/VariationalDynamics.lean`, 2025–2026.

[5] R. H. Coase, "The Nature of the Firm," *Economica*, 4(16):386–405, 1937.

[6] F. A. Hayek, "The Use of Knowledge in Society," *American Economic Review*, 35(4):519–530, 1945.

[7] L. von Mises, "Die Wirtschaftsrechnung im sozialistischen Gemeinwesen," *Archiv für Sozialwissenschaft und Sozialpolitik*, 47:86–121, 1920.

[8] V. Pareto, *Cours d'économie politique*, Lausanne, 1897.

[9] R. I. M. Dunbar, "Neocortex size as a constraint on group size in primates," *J. Human Evolution*, 22(6):469–493, 1992.

[10] P. Turchin, *Historical Dynamics: Why States Rise and Fall*, Princeton University Press, 2003.

[11] G. K. Zipf, *Human Behavior and the Principle of Least Effort*, Addison-Wesley, 1949.